Recap: Lecture 25

In Lecture 25, we

- Learnt how to show a series solution (from SoV) is genuine, for an example involving the 1D heat equation
- Saw that the heat equation smoothing creates genuine solutions even with challenging ICs
- Briefly discussed weak solutions (not assessed)

Lecture 26

In Lecture 26, we will:

1. Introduce the wave equation
2. Apply SoV to the 1D wave equation

The wave equation

The wave equation (a second order, linear, hyperbolic PDE)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u \qquad \text{(with constant velocity of propagation } c\text{)}$$

and its variations appear in many applications:

- Musical instruments: guitar strings, drum lids, etc.
- Acoustics: speech, music, sonar, etc.
- Shallow water waves: tsunamis, wave energy, coastal protection, etc.
- Waves in elastic bodies: earthquakes, structural vibrations, etc.
- Electromagnetic and gravitational waves: optics, X-rays, lasers, microwaves, etc
- Elementary particle physics: Quantum field theory and string theory

Wave equation: Problem 1

Consider a guitar string that is pinned at both ends and plucked at its centre.

Governing PDE: 1D wave equation $u(x,t)$ is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{for} \quad 0 < x < L \quad \text{and} \quad t > 0$$

Initial conditions:

$$u(x,0) = f(x) = \begin{cases} x & 0 \leqslant x < \frac{L}{2} \\ L - x & \frac{L}{2} \leqslant x < L \end{cases} \quad \text{for} \quad 0 < x < L$$

and

$$\dot{u}(x,0) = 0 \quad \text{for} \quad 0 < x < L$$

Boundary conditions: Dirichlet BCs

$$u(0,t) = u(L,t) = 0 \quad \text{for} \quad t > 0$$

Solving the 1D wave equation using separation of variables

1. Separate variables to turn single PDE into multiple ODEs

Use $u(x,t) = X(x)\,T(t)$ in the 1D heat equation to derive the ODEs

$$\ddot{T} - c^2\,\lambda\,T = 0 \quad \text{and} \quad X'' - \lambda\,X = 0$$

where λ is the separation constant.

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions.

As for the 1D heat equation Problem 1, we have

$$u(0,t) = X(0)\,T(t) = 0 \quad \Rightarrow \quad X(0) = 0$$

and

$$u(L,t) = X(L)\,T(t) = 0 \quad \Rightarrow \quad X(L) = 0$$

The eigenvalue problem is

$$X'' - \lambda\, X = 0 \quad \text{with} \quad X(0) = X(L) = 0.$$

This is identical to the eigenvalue problem in the 1D heat equation Problem 1! Therefore, the eigenvalues, $X_n(x)$, and associated eigenvalues, λ_n, are

$$X_n(x) = \sin(k_n\, x) \quad \text{and} \quad \lambda_n = -k_n^2 \quad \text{where} \quad k_n = \frac{n\,\pi}{L}$$

for $n = 1, 2, \ldots$ (note that $n = 0$ is not permitted)

3. Solve the remaining ODEs for each eigenvalue.

Let

$$\ddot{T}_n - c^2 \lambda_n T_n = 0 \quad \Rightarrow \quad T_n(t) = a_n \cos(ck_n t) + b_n \sin(ck_n t)$$

for $n = 1, 2, \ldots$ (note again that $n = 0$ is not permitted)

Note: We define $\omega_n = ck_n$ to make clear that the argument involves frequencies

4. Apply superposition principle to get a general solution in series form

$$u(x,t) = \sum_{n=1}^{\infty} \{a_n \cos(\omega_n t) + b_n \sin(\omega_n t)\} \sin(k_n x)$$

Remember that k_n and ω_n are linked through $\omega_n = ck_n$

5. Apply the initial conditions to find the series coefficients.

$$u(x,t) = \sum_{n=1}^{\infty} \left\{ a_n \cos(\omega_n t) + b_n \sin(\omega_n t) \right\} \sin(k_n x)$$

$$\Rightarrow \quad \dot{u}(x,t) = \sum_{n=1}^{\infty} \omega_n \left\{ -a_n \sin(\omega_n t) + b_n \cos(\omega_n t) \right\} \sin(k_n x)$$

We have two ICs

$$u(x,0) = \sum_{n=1}^{\infty} a_n \sin(k_n x) = f(x) \qquad (0 < x < L)$$

and

$$\dot{u}(x,0) = \sum_{n=1}^{\infty} \omega_n b_n \sin(k_n x) = 0 \qquad (0 < x < L)$$

i.e., two Fourier series for the odd periodic extensions of $f(x)$ and 0

Thus, by comparison of coefficients (or using orthogonal projections),

$$b_n = 0 \qquad \text{and} \qquad a_n = \frac{2}{L}\int_0^L f(x)\,\sin(k_n x)\,\mathrm{d}x$$

For our chosen IC

$$a_n = \frac{2}{L}\int_0^{L/2} x\,\sin(k_n x)\,\mathrm{d}x + \frac{2}{L}\int_{L/2}^{L} (L-x)\,\sin(k_n x)\,\mathrm{d}x$$

which we have already evaluated in Ex 4.9 (cf. 1D heat equation Problem 1), so that

$$a_n = \frac{4\,L}{n^2\pi^2}\sin(n\pi/2)$$

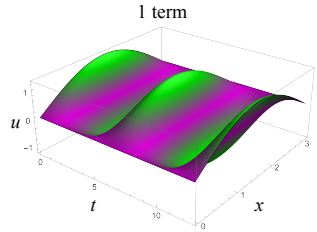
Therefore

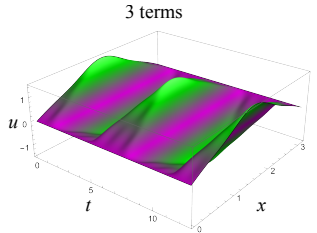
$$u(x,t) = -\frac{4\,L}{\pi^2}\sum_{m=1}^{\infty}\frac{(-1)^m}{(2m-1)^2}\,\cos(\omega_{2m-1}t)\,\sin(k_{2m-1}x)$$

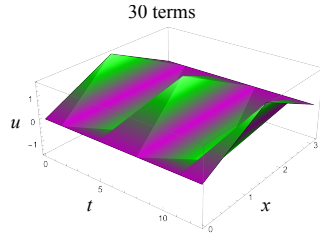
Compare with the corresponding solution for the 1D heat equation

$$u(x,t) = -\frac{4L}{\pi^2}\sum_{m=1}^{\infty}\frac{(-1)^m}{(2m-1)^2}\mathrm{e}^{-\alpha k_{2m-1}^2 t}\sin(k_{2m-1}x)$$

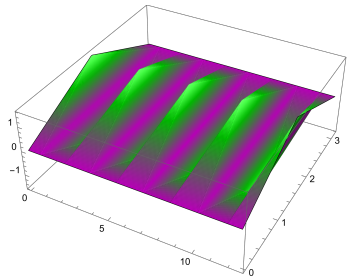
Plotting the results with $L = \pi$ and $c = 1$



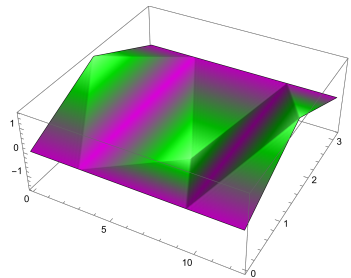




$c = 2$: wave propagates twice as fast



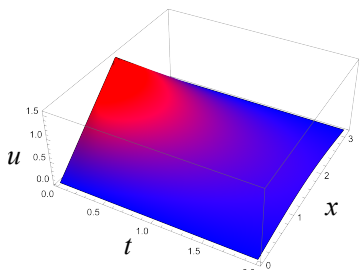
$c = 1/2$: halves the speed of the wave

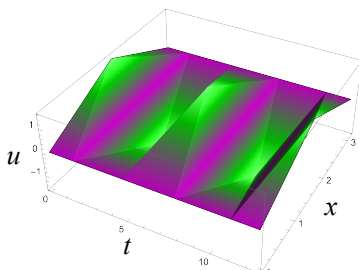


Compare heat and wave operators

1. Heat equation smooths out the discontinuities and kinks in the initial condition.

2. Wave equation propagates the discontinuities and kinks in the initial condition periodically in time.





Discussion: Our formal solution

$$u(x,t) = -\frac{4\,L}{\pi^2} \sum_{m=1}^{\infty} \frac{(-1)^m}{(2m-1)^2} \cos(\omega_{2m-1} t) \sin(k_{2m-1} x)$$

Genuine solution?

- Each term in the solution repeats every $t = 2nL/c$
- Can the solution satisfy the 1D wave equation throughout the domain?
- This is a weak solution

Steady state?

- What happens as $t \to \infty$? Can we define a steady state?